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Course Code:DSA0112

Slot-C

QUESTIONS 9 / 9 completed

Reverse

REVERSE

Write a function to reverse a given string.

PROGRAM EDITOR

C++

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     string s;
5     cin >> s;
6     for (int i=s.length()-1;i>=0;i--){
7         cout << s[i];
8     }
9     return 0;
10 }
```

OUTPUT

olleh

INPUT

hello

QUESTIONS 9 / 9 completed

GCD-Fun. ▾

GCD-FUN.

Write a function to determine the GCD (greatest common divisor) of two given integers.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int gcd(int a,int b){
4     while (b!=0){
5         int temp = b;
6         b=a%b;
7         a=temp;
8     }
9     return a;
10 }
11 int main()
12 {
13     int a,b;
14     cin >> a >> b;
15     cout << gcd(a,b);
16     return 0;
17 }
```

OUTPUT

6

INPUT

12 18

QUESTIONS 9 / 9 completed

Prime-Fun.

PRIME-FUN.

Write a function to determine if a given integer is a prime number or not.

PROGRAM EDITOR

C++

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int n;
5     cin >> n;
6     bool prime = true;
7     if (n <= 1)
8         prime = false;
9     else{
10         for (int i=2;i<=n;i++){
11             if (n%i==0){
12                 prime = false;
13                 break;
14             }
15         }
16     }
17     if(prime)
18         cout << "prime";
19     else
20         cout << "not prime";
21     return 0;
```

OUTPUT

not prime

INPUT

6

QUESTIONS 9 / 9 completed

Factorial-Fun.

FACTORIAL-FUN.

Write a function to compute the factorial of a given integer.

PROGRAM EDITOR

C++

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 long long factorial(int n){
4     if(n<0)
5         return -1;
6     long long result = 1;
7     for (int i=2;i<=n;i++){
8         result *=i;
9     }
10    return result;
11 }
12 int main(){
13     int n;
14     cin >> n;
15     cout << factorial(n);
16     return 0;
17 }
```

OUTPUT

120

INPUT

5

QUESTIONS 9 / 9 completed

Count

COUNT

Write a function to count the number of words in a given string.

PROGRAM EDITOR

C++

Run

Save

```
1 #include <bits/stdc++.h>
2 using namespace std;
3 int countWords(const string& s) {
4     int count = 0;
5     bool inWord = false;
6
7     for (char c : s) {
8         if (isspace((unsigned char)c)) {
9             inWord = false;
10        } else {
11            if (!inWord) {
12                count++;
13                inWord = true;
14            }
15        }
16    }
17    return count;
18 }
19
20 int main() {
21     string str;
```

OUTPUT

5

INPUT

WRITE A FUNCTION
TO COUNT